

Mass Flow & Humidity



Ultrasonic Flowmeters

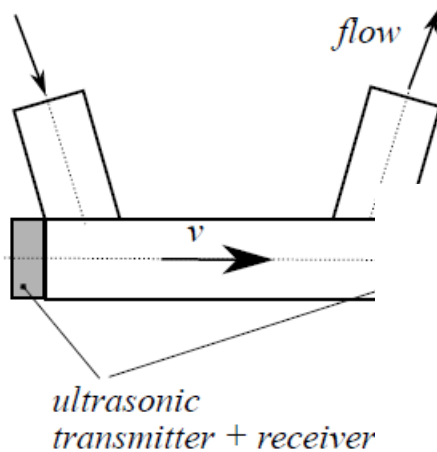


FIGURE 12.16: Principle of ultrasonic

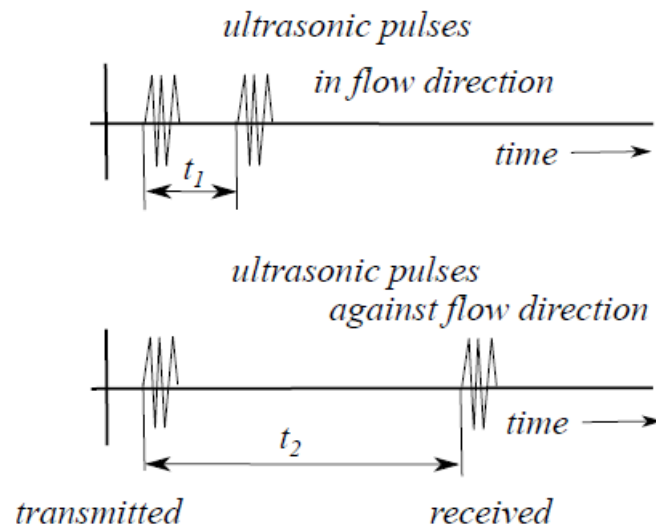


FIGURE 12.17: Timing of ultrasonic signals

Electromagnetic Flowmeters

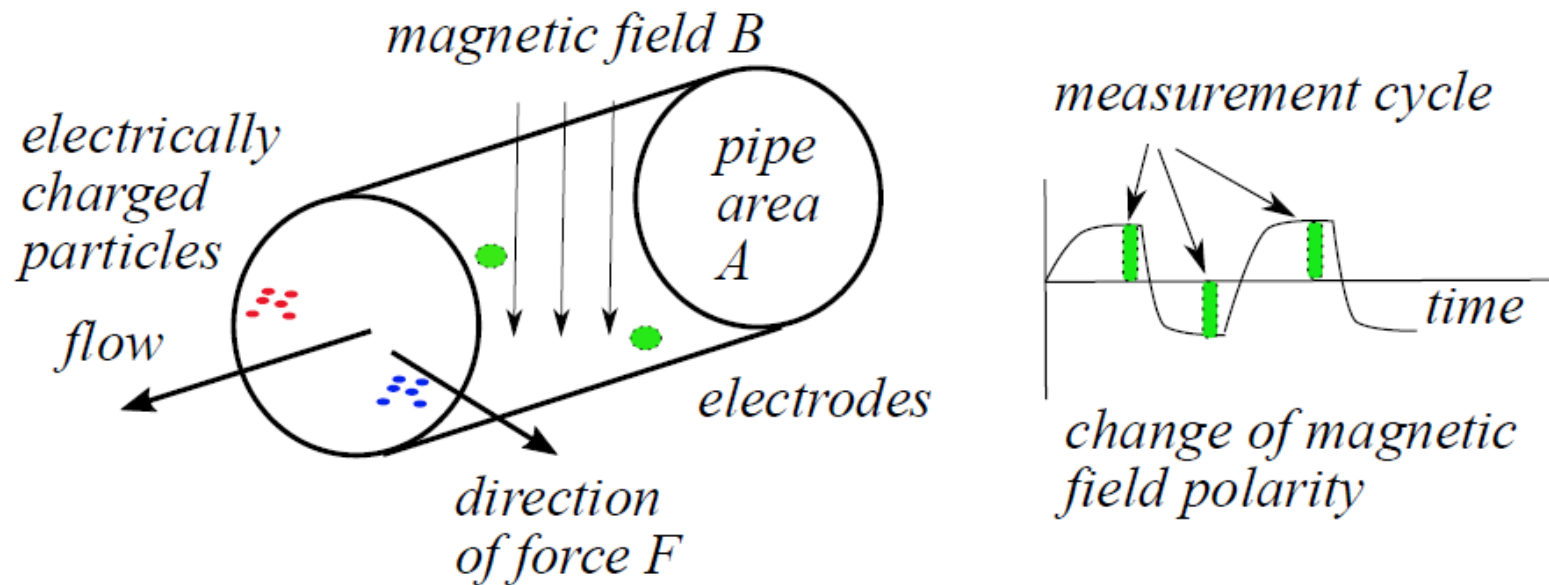
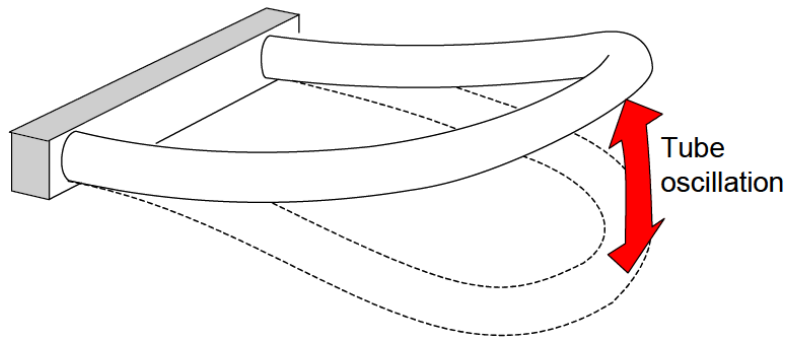
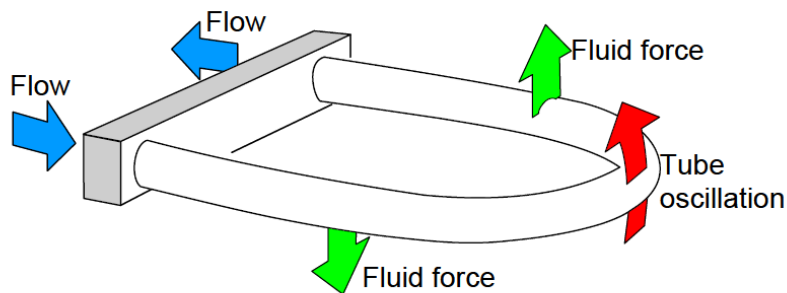


FIGURE 12.22: Principle of electromagnetic flowmeter

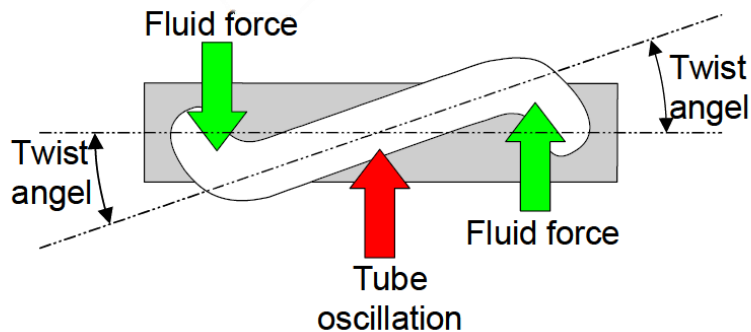
Coriolis Flowmeters



Oscillatory motion applied to a single tube

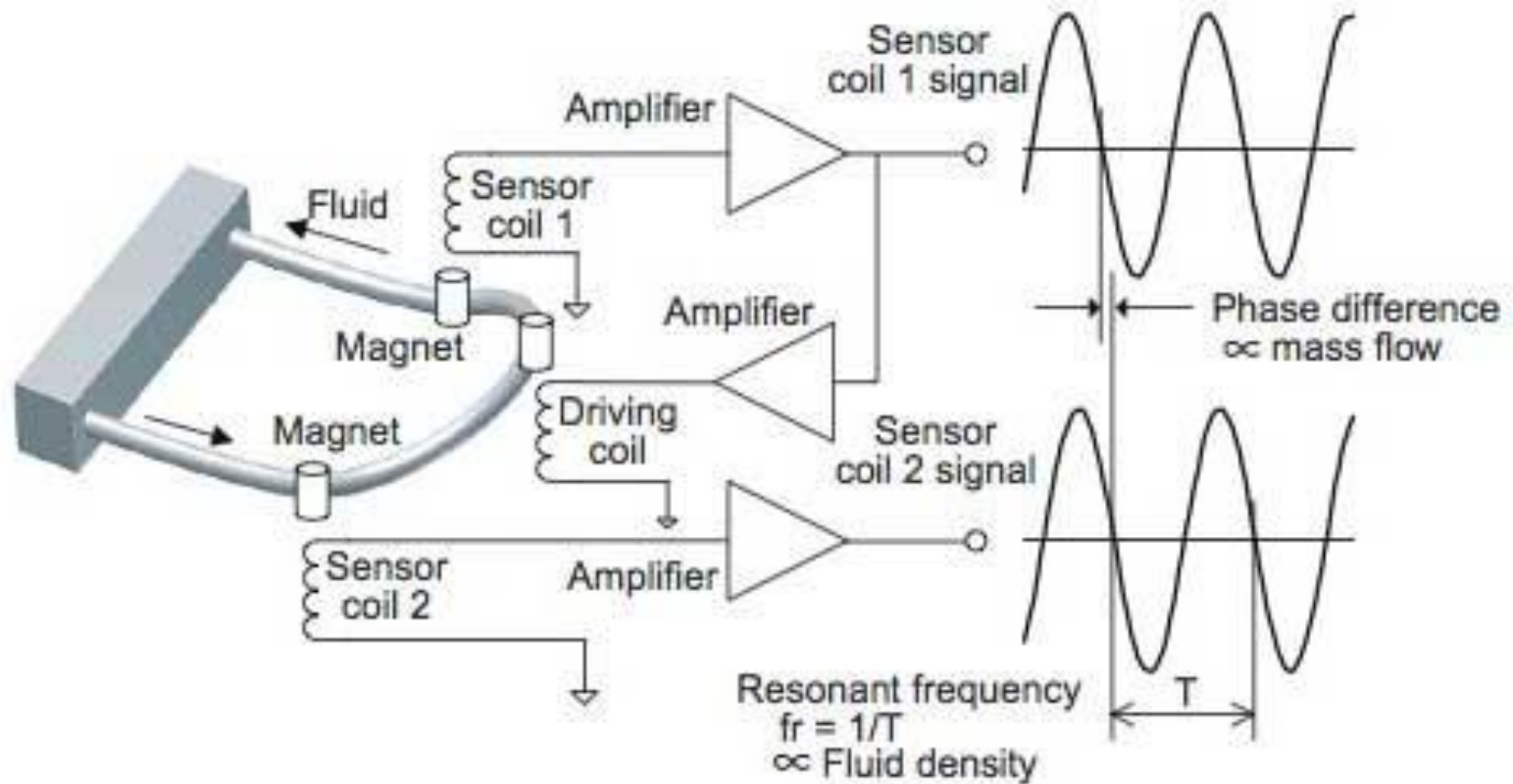


Forces acting on the tube as a result of fluid flow



The complete loop is twisted by an amount that is directly and linearly to the mass flow rate of the fluid

Coriolis Flowmeters



Hygrometers with dry electrolytes

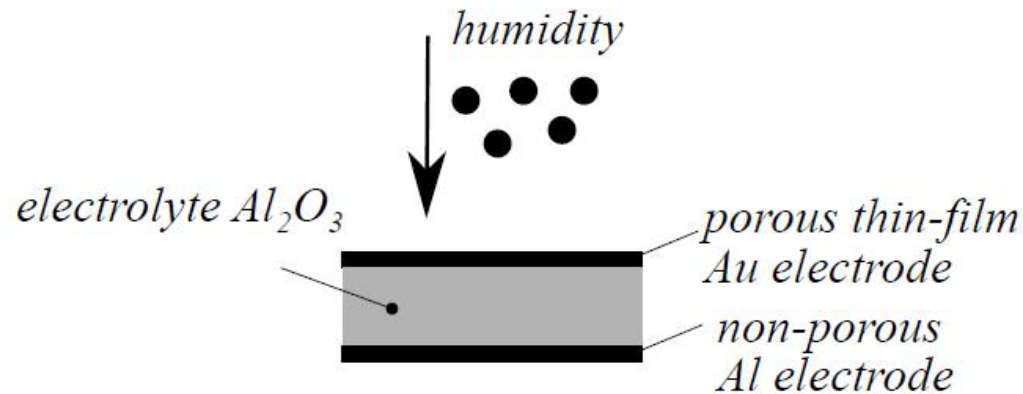
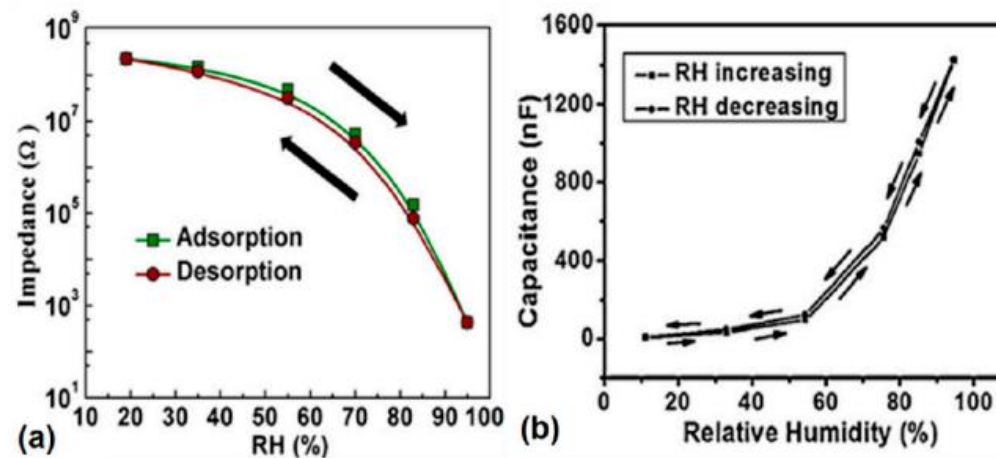


FIGURE 11.6: Hygrometer with dry electrolyte



Hygrometers with dry electrolytes

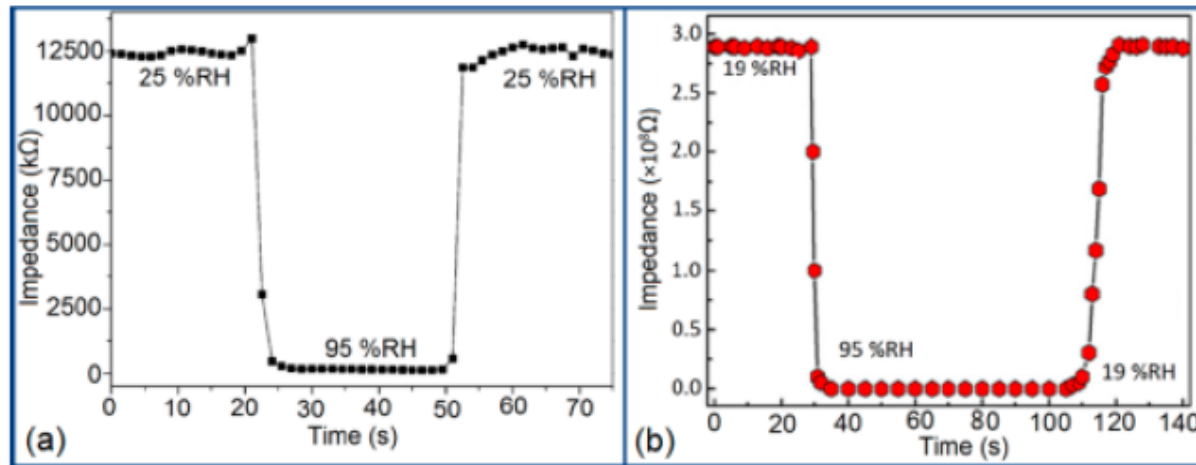


FIGURE 11.8: Transient response of a humidity sensor for various materials, a) KCl doped ZnO nanofiber, b) ZnO, courtesy of [68], license CC BY 3.0

Hygrometers with dry electrolytes

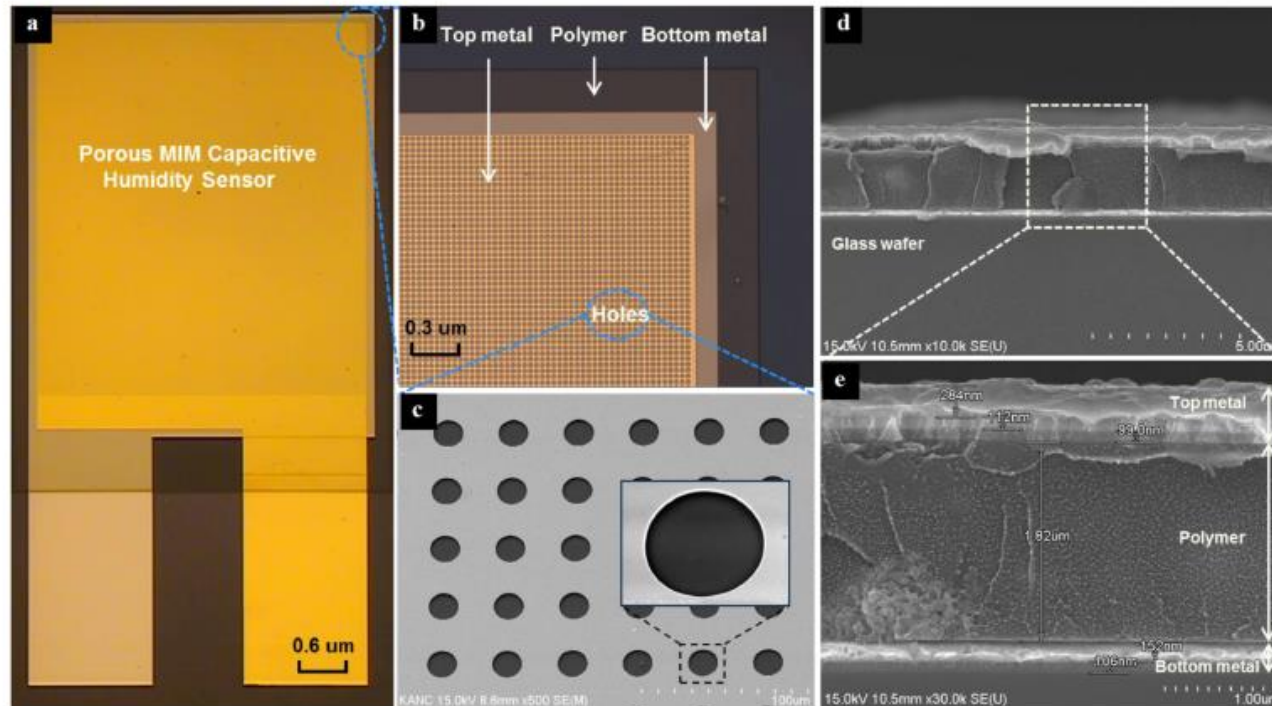
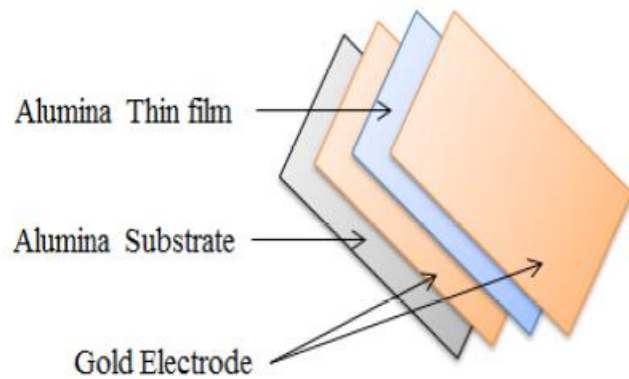
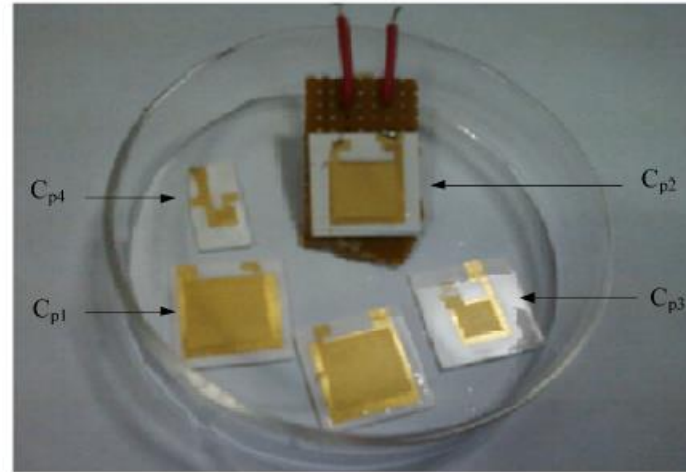


FIGURE 11.9: a) Capacitive humidity sensor, b) detail, c) detail of porous structure, d) cut through the sensor, courtesy of [69], license CC BY 4.0

Hygrometers with dry electrolytes



(a)



(b)

FIGURE 11.10: Capacitive humidity sensor, courtesy of [70], license CC



FIGURE 11.11: Humidity sensor,
removed plastic cover—photo
author



FIGURE 11.12: Humidity
sensors—photo author